

Faculty of Electrical - Electronics Engineering
Department of Control and Automation Engineering
KOM4531 - Adv. Industrial Automation Sys. Applications
Experiment 1 and 2
MODBUS Communication Experiment

In this document, projects and their guidance for each automation laboratory, which are Phoenix Contact and Schneider Electric, are given. You will learn how to create a new project, add devices and their modules, use necessary instructions to achieve MODBUS communication via writing Ladder diagram.

1 MODBUS TCP Experiment at Phoenix Contact Lab.

Every education set is equipped with Windows 7 OS which has **AUTOMATIONWORX Software Suite 1.89** program and a PLC. These devices are connected via an Ethernet hub.

You can visit PC WORX DEMO webpage to download the same version of the program to continue your project from your PC.

1.1 The Project

You are responsible from the creation of the server part on a PLC and client part on another PLC as an engineer to communicate them via MODBUS TCP. So you are going to use 2 PLCs during the experiment.

You need to obtain a value from the analog input on the server side. If the settings button on the server side is false, you need to assign this value to address 5. If the settings button on the Server side is true, you need to subtract this value from 32768 and assign the result to address 5. You should also assign the value of this button to the first address on the Server. The value of this button and the value of the analog input should be read from the Client side. Also; after the Client is connected to the server, the address (to which the analog input is assigned) is shifted by 10 for every 3 seconds and the value of 0 is assigned to the old address.

If you choose not to do the shifting part, the grading will be out of 70.

Starting procedure of this project is given as below:

1. Create a new project in PC-Worx.
2. From Project Tree Window (docked at left side), right-click **Libraries** and left-click **Insert** and **User Library** respectively.
3. Select **Modbus_TCP_2.mwt** from the folder in desktop.
4. Open the Main POU and check that you are able to select **Modbus_TCP_2** in the Function library.
5. Rebuild the empty project before starting it to make sure that there are no errors.

2 MODBUS RTU Experiment at Schneider Electric Lab.

Every education set is equipped with Windows 7 OS which has **TIA PORTAL V16 Update 7** program, a PLC, a motor driver and a motor. PLC and motor driver are connected via an RS-485 cable.

You can visit TIA PORTAL V16 trial, PLCSIM V16 and Update 7 to download the same version of program to continue your project from your PC.

Firmware versions of PLC and HMI used on your education set are given below:

Device	Model - Version
PLC	V2.2
HMI	KTP600 Basic mono PN

Table 1: Device Models and Versions

Serial communication settings for motor drivers are given below:

Baud Rate	Parity
9600	Even (8E1)

Table 2: Serial Communication Settings

2.1 The Project

As an engineer, you are responsible from the communication between the motor driver and Siemens PLC to drive the motor. Every education set is equipped with different brand and model of motor drivers which are given in Table 3. Motor driver and PLC will be communicated via MODBUS RTU.

Note that you are not allowed to change any motor driver settings from the motor driver. Hence, it is configured for you and it is tested that it is working. If you change any settings from the motor driver, you can't finish your project.

Brand	Model
Delta	VFD-EL-W
PowerFlex	4M
Schneider	Altivar 31
Schneider	Altivar 32

Table 3: Available Motor Drivers

You need to complete 3 main objectives given below to finish the project:

1. Establish a proper connection between motor driver and PLC.
2. Design an HMI page to change driving status and speed.
3. Design an algorithm to achieve the desired fail-safe mechanism and write it as a Ladder diagram.

Communication	
Name	Description
▶ S7 communication	
▶ Open user communication...	
▶ WEB Server	
▶ Others	
▼ Communication processor	
▶ PtP Communication	
▶ USS communication	
▶ MODBUS (RTU)	
▶ Point-to-point	
▶ USS	
▼ MODBUS	
■ MB_COMM_LOAD	Configure port on the PtP module for Modbus RTU
■ MB_MASTER	Communicate via the PtP port as Modbus master
■ MB_SLAVE	Communicate via the PtP port as Modbus slave

Figure 1: MODBUS RTU Instructions

2.1.1 Creating New Project

You are going to create a new project and add devices given in Table 1. Note that you need to figure the PLC model out by looking the PLC case. Also, you need to add the communication module to your PLC from Device Configuration. You can see the module number of the communication module by looking the module case.

2.1.2 Establishing Communication

You are going to use MODBUS RTU instructions to establish a connection between PLC and motor driver. These instructions can be found from Communication tab. Left-click **Communication processor** and **MODBUS** folders to list available MODBUS RTU instructions you may use in your program, see Figure 1. Here are few tricks about programming to be noted:

- Your program must use 1 **MB_COMM_LOAD** instruction.
- Your program must use at least 2 **MB_MASTER** or **MB_SLAVE** instructions. You need to figure it out which one to be used. The number of instruction depends on how you constructed your algorithm, but 2 or 3 instructions are expected to be used.
- If you are going to use data blocks, make sure that data blocks must allow direct addressing. When a data block is created, standard access attribute must be selected via unticking **Optimized block access** in **Attributes** tab from its properties to achieve this.

For additional information, you can press F1 after left-clicking an instruction to show its help page from the information system.

See **Section 13.9** from S7-1200 System Manual for more detailed information with examples, either in English or Turkish.

Here are few tricks about motor drive to be noted:

- You must look for keywords such that *MODBUS*, *MODBUS address*, *communication*, *control word*, *output frequency* in your driver's datasheet.
- Motor drivers can be in the error state if there is a time-out on the communication line. You must clear the error state to operate the motor drive again. See control word of your motor driver datasheet.
- If you have configured serial communication settings properly, you should see some indications on the motor driver while writing. These indications can be direction lamp and commanded speed changed on the motor driver screen.

2.1.3 Design of HMI Page

You are going to design an HMI page to allow the operator to drive the motor with configured direction and speed settings. An HMI page similar to Figure 2 must be designed.

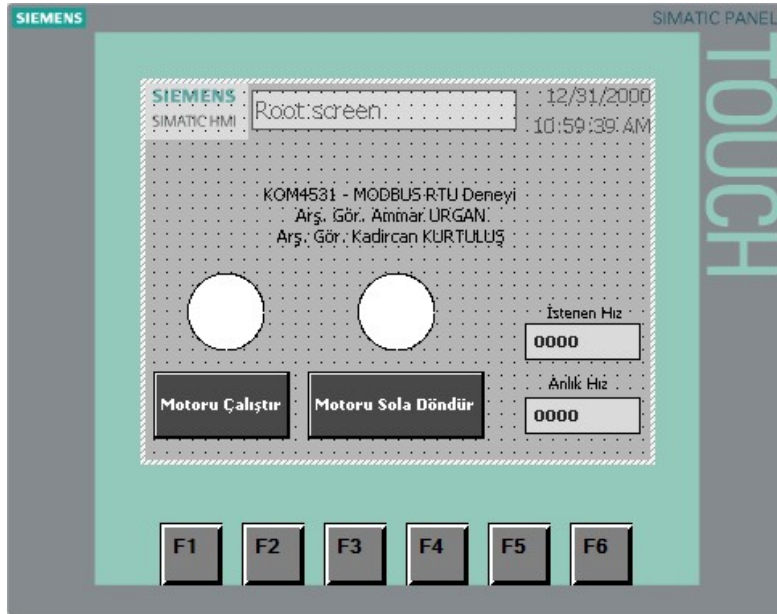


Figure 2: Desired HMI page to be designed

HMI page must include these basic objects:

- **2 Circles** which indicate motor start/stop and direction buttons from education set.
 - You should add a display animation from their properties. Circle colors must be
 - * red if the corresponding education set button is false.
 - * green if the corresponding education set button is true.
- **2 Text Fields** above I/O fields which indicate texts of commanded and output speed.
- **A Text Field** on top of the page which indicates working group information (group number - student names and surnames).

HMI page must include these elements:

- **2 I/O Fields** which indicate values of commanded and output speed.
 1. First one is going to be used as input to enter the desired speed and the value should be sent to the motor driver.
 2. Last one is going to be used as output and it will show the output speed read from the motor driver.
- **2 Switches** which allow the operator to start/stop the motor and change its direction. Note that these buttons will be a part of a fail-safe algorithm on PLC programming side.

2.1.4 Design of Fail-Safe Algorithm

You are going to design an algorithm to prevent the operator drive the motor without authorization. This algorithm requires 2 education set buttons to be used:

- First button will authorize motor start/stop process. Operator can start/stop the motor driver if this button is true.
 - This means that motor will start if corresponding buttons on HMI and PLC are true.
- Last button will authorize changing motor direction. Operator can change the direction if this button is true.
 - This means that the motor will change direction if corresponding buttons on HMI and PLC are true.

2.2 Grading

- Every design criteria will be graded, percentage is the highest and TBA.
- Attendance to the laboratory is a must. Also, it will affect your grade. Percentage is lower and TBA.
- If you are able to finish your project in the first week, bonus points can be given (TBD).
- You will be asked questions about the project. Also, answers of your teammates can increase/decrease your own grade. Percentage is higher and TBA.

2.3 Questions

You are not going to get answers for every question you asked and you must be prepared before the laboratory. But if you have programming problems related to TIA Portal and unable to continue, you can ask the assistant freely. Additionally, the assistant can give some hints and tricks about programming and your motor driver after some time passed in the laboratory.